

Literature Answer: c_0 Does Not Embed Into $\mathcal{F}(\ell_2)$

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Status. `literature_already_answered`. The extracted open problem from arXiv:1505.07209 has a negative answer in later literature.

Source question. In Cuth–Doucha–Wojtaszczyk, *On the structure of Lipschitz-free spaces*, arXiv:1505.07209, Section “Open problems” asks, after Theorem 4.1 on finite-dimensional Euclidean subsets,

Is it true that $c_0 \hookrightarrow \mathcal{F}(\ell_2)$?

In the source file this is the unnumbered question at line 349. The later supporting paper refers to it as Question 3 of Cuth–Doucha–Wojtaszczyk.

Later answer. Aliaga–Nous–Petitjean–Prochazka, *Compact reduction in Lipschitz free spaces*, arXiv:2004.14250, proves compact reduction for weak sequential completeness. Corollary 2.2 states that if X is superreflexive, then $\mathcal{F}(X)$ is weakly sequentially complete. Immediately after the corollary the authors write that, in particular, $\mathcal{F}(\ell_2)$ is weakly sequentially complete and that this provides a negative answer to Question 3 posed by Cuth–Doucha–Wojtaszczyk.

Why this answers the source question. Weak sequential completeness passes to closed subspaces. The space c_0 is not weakly sequentially complete: the sequence of partial sums $(e_1 + \dots + e_n)_n$ is weakly Cauchy in c_0 , since each $\ell_1 = c_0^*$ functional sees convergent scalar partial sums, but it has no weak limit in c_0 . Therefore a weakly sequentially complete Banach space cannot contain an isomorphic copy of c_0 . Since $\mathcal{F}(\ell_2)$ is WSC by arXiv:2004.14250, one has

$$c_0 \not\hookrightarrow \mathcal{F}(\ell_2).$$

Scope limitations. This is a full negative literature answer for the ℓ_2 question from arXiv:1505.07209. It does not settle the adjacent source question asking whether $c_0 \hookrightarrow \mathcal{F}(\ell_1)$.

Search evidence. Cheap index search found no prior exact packet for arXiv:1505.07209. Local corpus searches for c_0 , $\mathcal{F}(\ell_2)$, weak sequential completeness, and Lipschitz-free spaces surfaced arXiv:2004.14250, which explicitly names and answers the source question. Web searches for the exact c_0 and $\mathcal{F}(\ell_2)$ phrases did not surface a newer contradictory status.

References.

- M. Cuth, M. Doucha, and P. Wojtaszczyk, *On the structure of Lipschitz-free spaces*, arXiv:1505.07209.

- R. J. Aliaga, C. Nous, C. Petitjean, and A. Prochazka, *Compact reduction in Lipschitz free spaces*, arXiv:2004.14250.